Table 1: Secondary threshold-based performance in the nonlinear simulation extension. Values are means over 100 repetitions under the common same-model prevalence-quantile decision rule. Balanced accuracy, F1, and Matthews correlation coefficient (MCC) are secondary threshold-dependent measures. In the one-sample design the threshold is estimated from the full validation population; in the two-sample design it is estimated from validation U only. The selected training-sample model is used both to determine the validation threshold and to score the test sample.

One-sample - Additive sinusoidal				
Method	Balanced accuracy	F1	MCC	n
Logistic probability correction	0.6147	0.3838	0.2297	100
nnPU logistic	0.6150	0.3843	0.2303	100
Naive CatBoost	0.6464	0.4345	0.2930	100
RN-CatBoost	0.6287	0.4062	0.2575	100
CS-CatBoost	0.6498	0.4401	0.3000	100
PU gradient boosting	0.6454	0.4322	0.2882	100

One-sample - Radial / concentric				
Method	Balanced accuracy	F1	MCC	n
Logistic probability correction	0.5289	0.2462	0.0579	100
nnPU logistic	0.5291	0.2465	0.0583	100
Naive CatBoost	0.7230	0.5568	0.4461	100
RN-CatBoost	0.5015	0.2018	0.0029	100
CS-CatBoost	0.7233	0.5573	0.4467	100
PU gradient boosting	0.7066	0.5296	0.4093	100

One-sample - Pure XOR interaction				
Method	Balanced accuracy	F1	MCC	n
Logistic probability correction	0.5174	0.2276	0.0348	100
nnPU logistic	0.5173	0.2274	0.0346	100
Naive CatBoost	0.7172	0.5474	0.4345	100
RN-CatBoost	0.5361	0.2577	0.0722	100
CS-CatBoost	0.7179	0.5486	0.4359	100
PU gradient boosting	0.5698	0.3160	0.1376	100

One-sample - Threshold interaction				
Method	Balanced accuracy	F1	MCC	n
Logistic probability correction	0.5008	0.2011	0.0017	100
nnPU logistic	0.5008	0.2011	0.0016	100
Naive CatBoost	0.5659	0.3049	0.1320	100
RN-CatBoost	0.5026	0.2040	0.0052	100
CS-CatBoost	0.5771	0.3230	0.1543	100
PU gradient boosting	0.5003	0.2358	0.0005	100

One-sample - Sinusoidal + 497 noise predictors ($p = 500$)

Method	Balanced accuracy	F1	MCC	n
Logistic probability correction	0.5419	0.2674	0.0838	100
nnPU logistic	0.5509	0.2816	0.1017	100
Naive CatBoost	0.6257	0.4012	0.2516	100
RN-CatBoost	0.6237	0.3979	0.2478	100
CS-CatBoost	0.6260	0.4017	0.2521	100
PU gradient boosting	0.6361	0.4135	0.2583	100

One-sample - Nonlinear heterogeneous covariance

Method	Balanced accuracy	F1	MCC	n
Logistic probability correction	0.5086	0.2140	0.0172	100
nnPU logistic	0.5084	0.2137	0.0167	100
Naive CatBoost	0.6895	0.5030	0.3789	100
RN-CatBoost	0.5172	0.2275	0.0345	100
CS-CatBoost	0.6926	0.5079	0.3849	100
PU gradient boosting	0.5559	0.2950	0.1093	100

Two-sample - Additive sinusoidal

Method	Balanced accuracy	F1	MCC	n
Logistic probability correction	0.6161	0.3853	0.2317	100
nnPU logistic	0.6164	0.3858	0.2322	100
Naive CatBoost	0.6461	0.4333	0.2917	100
RN-CatBoost	0.6284	0.4051	0.2566	100
CS-CatBoost	0.6493	0.4383	0.2980	100
PU gradient boosting	0.6471	0.4341	0.2906	100

Two-sample - Radial / concentric

Method	Balanced accuracy	F1	MCC	n
Logistic probability correction	0.5228	0.2364	0.0455	100
nnPU logistic	0.5251	0.2401	0.0501	100
Naive CatBoost	0.7231	0.5567	0.4453	100
RN-CatBoost	0.5030	0.2051	0.0060	100
CS-CatBoost	0.7230	0.5566	0.4453	100
PU gradient boosting	0.7062	0.5319	0.4024	100

Two-sample - Pure XOR interaction

Method	Balanced accuracy	F1	MCC	n
Logistic probability correction	0.5175	0.2276	0.0349	100
nnPU logistic	0.5192	0.2304	0.0384	100
Naive CatBoost	0.7148	0.5431	0.4290	100
RN-CatBoost	0.5349	0.2557	0.0697	100
CS-CatBoost	0.7154	0.5441	0.4302	100
PU gradient boosting	0.5628	0.3080	0.1243	100

Two-sample - Threshold interaction

Method	Balanced accuracy	F1	MCC	n
Logistic probability correction	0.5014	0.2025	0.0027	100
nnPU logistic	0.5014	0.2026	0.0027	100
Naive CatBoost	0.5597	0.2961	0.1191	100
RN-CatBoost	0.5021	0.2041	0.0041	100
CS-CatBoost	0.5710	0.3140	0.1417	100
PU gradient boosting	0.5010	0.2313	0.0021	100

Two-sample - Sinusoidal + 497 noise predictors ($p = 500$)

Method	Balanced accuracy	F1	MCC	n
Logistic probability correction	0.5407	0.2650	0.0812	100
nnPU logistic	0.5472	0.2755	0.0942	100
Naive CatBoost	0.6222	0.3952	0.2439	100
RN-CatBoost	0.6199	0.3915	0.2389	100
CS-CatBoost	0.6226	0.3958	0.2444	100
PU gradient boosting	0.6366	0.4141	0.2586	100

Two-sample - Nonlinear heterogeneous covariance

Method	Balanced accuracy	F1	MCC	n
Logistic probability correction	0.5060	0.2091	0.0121	100
nnPU logistic	0.5067	0.2104	0.0135	100
Naive CatBoost	0.6806	0.4892	0.3617	100
RN-CatBoost	0.5193	0.2313	0.0387	100
CS-CatBoost	0.6846	0.4956	0.3699	100
PU gradient boosting	0.5524	0.2908	0.1026	100

Note. The six methods are prespecified fixed representatives. Corrected methods use raw source-membership scores. No numerical threshold is transferred to a refitted model. All 7,200 displayed method/repetition records are available; unavailable RN final refits do not affect these selected-model hard metrics.